

Predicting Glucose Levels

A Complete Data Science Pipeline

DATASET

Pima Indians Diabetes

RECORDS

768 patients · 9 features

COURSE

Data Analysis & Data Science

The Problem

Objective — Predict a patient's Glucose level from other physiological measurements using Linear Regression.

Dataset — 768 patient records: age, BMI, insulin, blood pressure, pregnancies, skin thickness, and diabetic status.

Target variable — Glucose (continuous, 44–199 mg/dL) — valid for Linear Regression.

Goal — End-to-end pipeline from raw data to a deployed interactive dashboard.

Our Methodology

1

Collection & Preprocessing

Identify and fix erroneous zero values in physiological columns.

2

Exploratory Data Analysis

Visualize distributions, correlations, and feature relationships.

3

Modeling

Apply Linear Regression to predict continuous Glucose levels.

4

Deployment

Build an interactive Streamlit web app with real-time prediction.

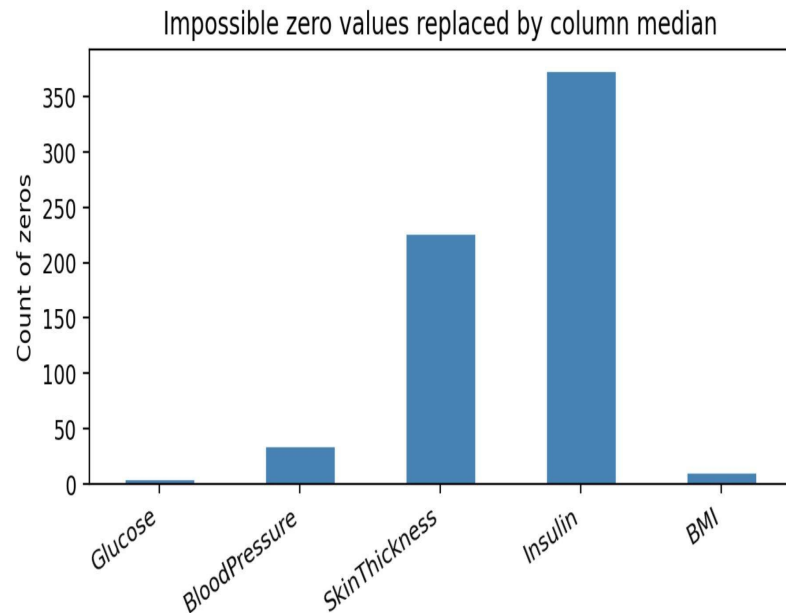
Data Preprocessing

The problem — BMI, BloodPressure, Insulin, SkinThickness, Glucose had impossible 0 values.

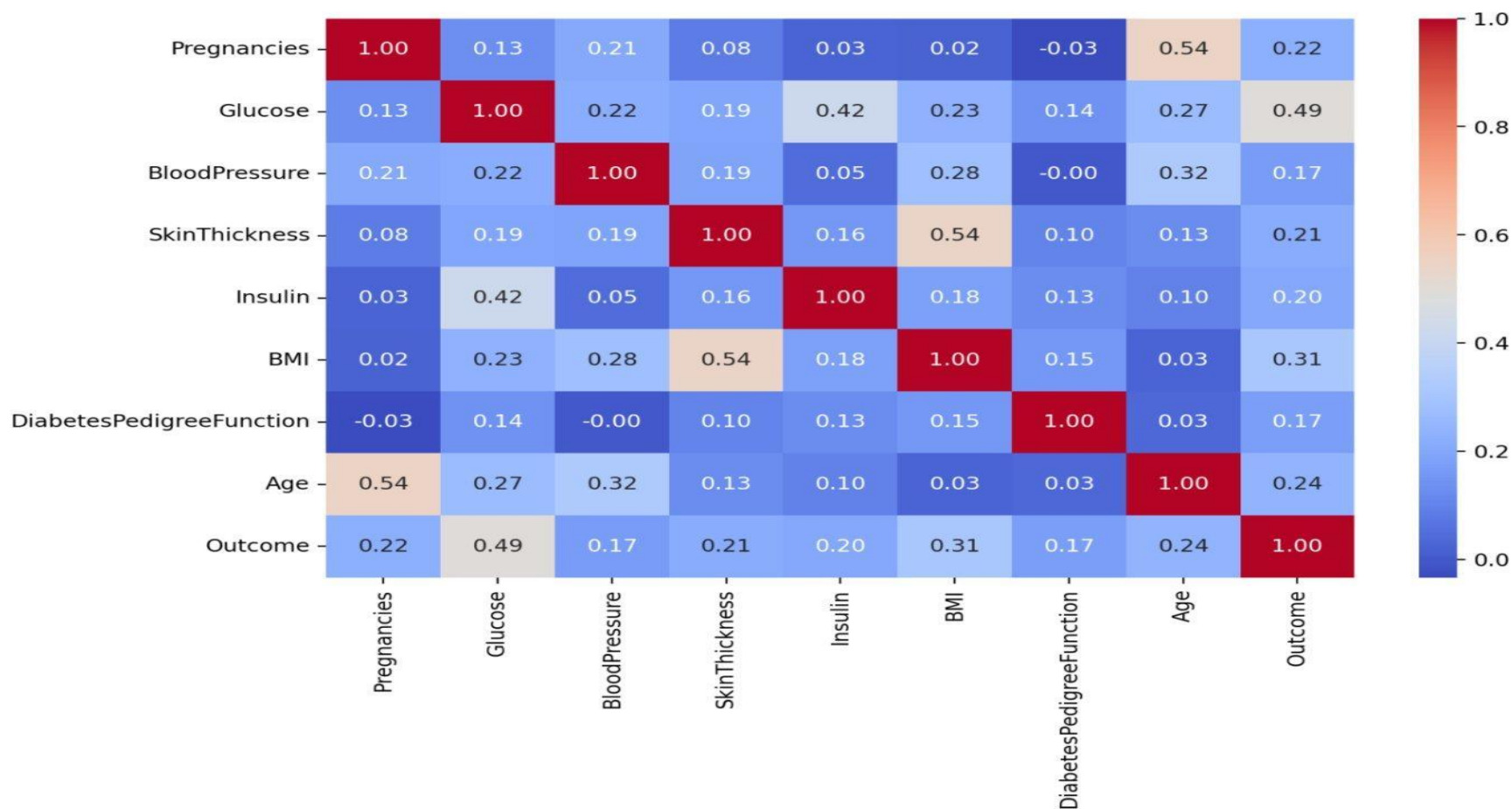
The fix — Zeros replaced with NaN, then imputed with column median.

Why median? — Resistant to outliers, unlike the mean.

Pregnancies left as-is — Zero is a valid value there.



Correlation Heatmap



EDA — Key Findings

Glucose ↔ Insulin: 0.42 — Strongest pairwise correlation, reflects insulin resistance.

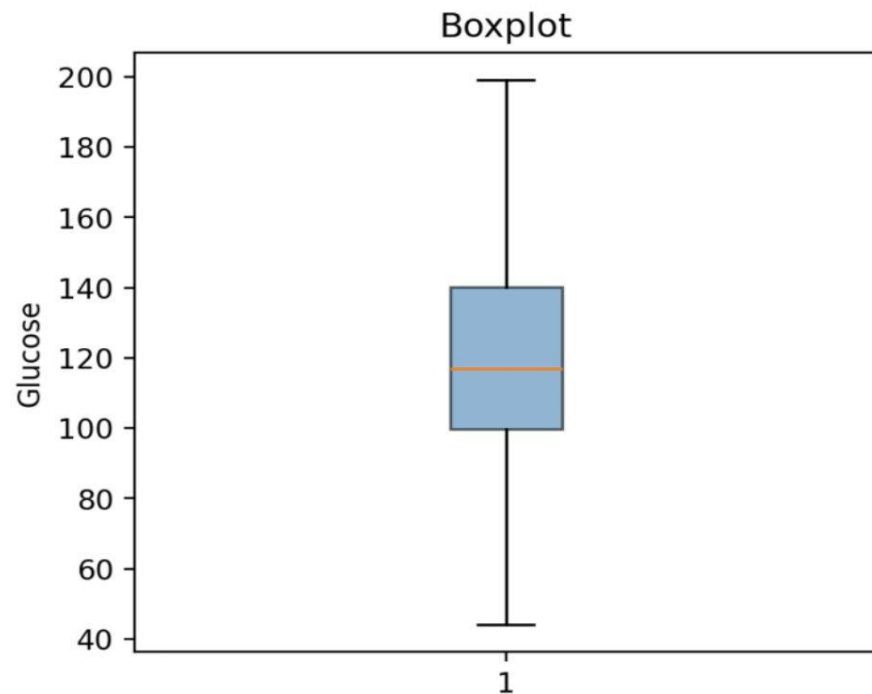
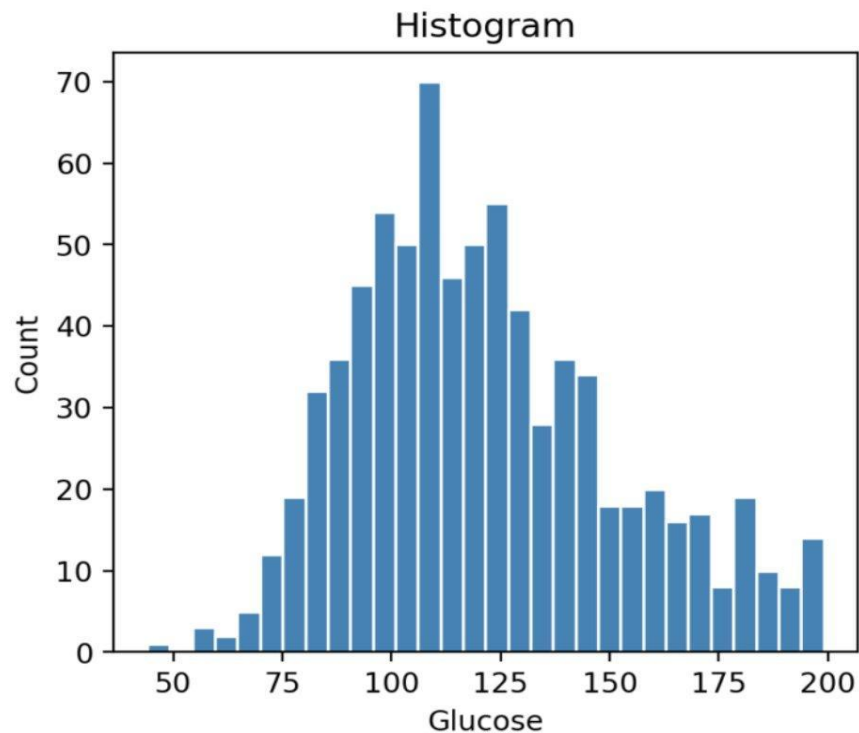
Glucose ↔ Outcome: 0.49 — Diabetic patients have higher glucose, as expected.

Glucose ↔ Age: 0.27 — Older patients trend toward higher glucose levels.

Glucose ↔ BMI: 0.23 — Higher body mass links to glucose dysregulation.

These correlations confirm Glucose can be reasonably predicted from the other features.

Glucose Distribution — Target Variable



Modeling — Setup

Target — Glucose (mg/dL) — continuous, valid regression target.

Features — Pregnancies, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Outcome.

Split — 80% training / 20% testing, random_state=42.

Scaling — StandardScaler applied so coefficients are comparable across features with different units.

Model — sklearn.linear_model.LinearRegression

Model Results

0.33

R² SCORE

672

MSE

~20

MAE (MG/DL)

~26

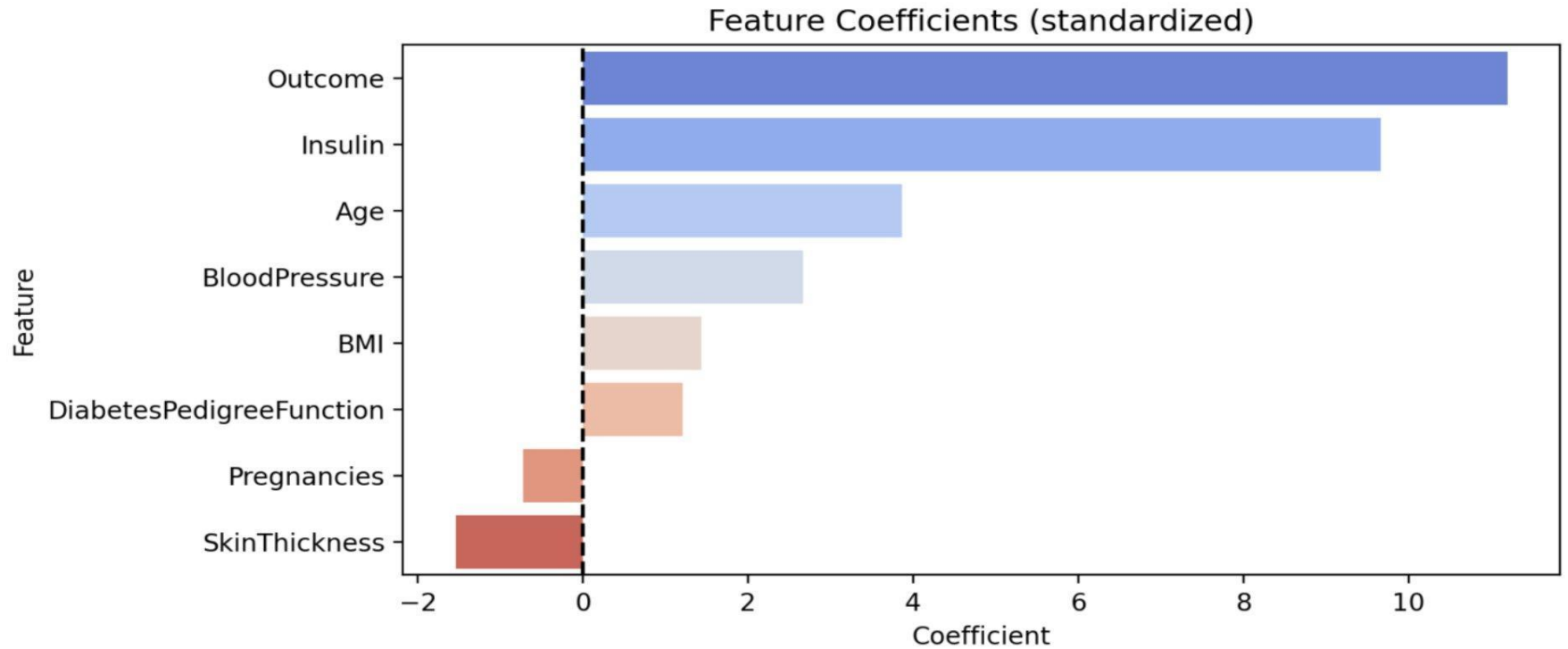
RMSE (MG/DL)

Outcome & Insulin — Strongest positive predictors by far.

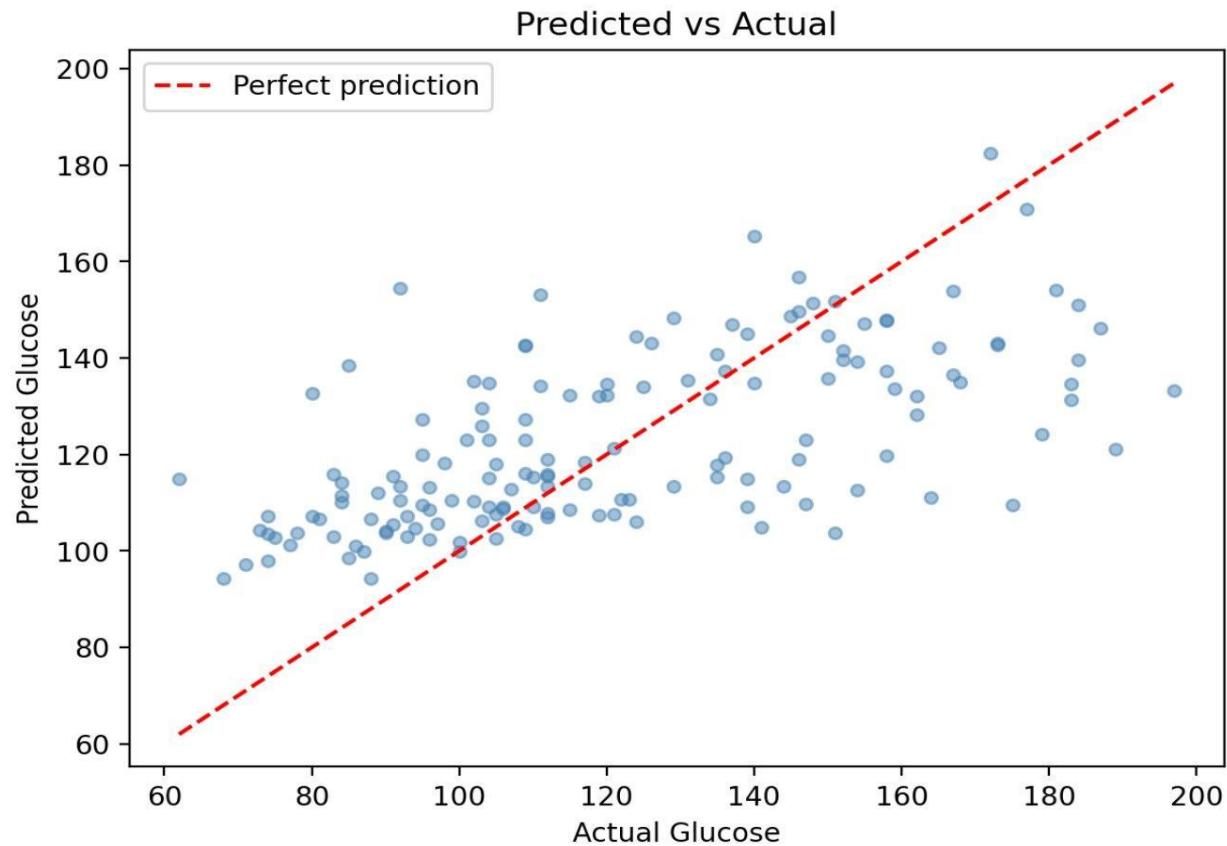
Age & BloodPressure — Moderate positive effect on Glucose.

SkinThickness & Pregnancies — Slight negative coefficients after standardization.

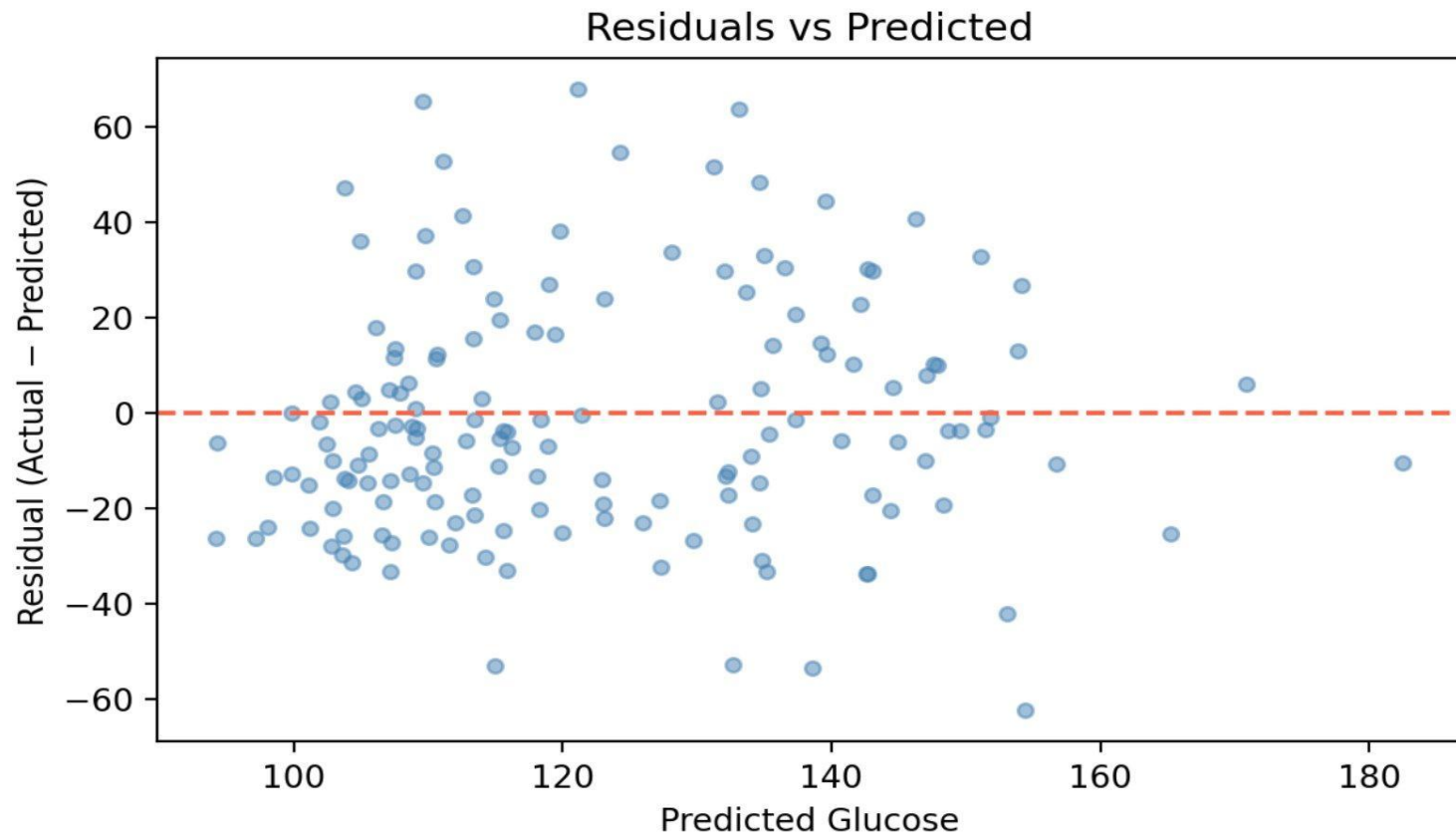
Feature Coefficients (standardized)



Predicted vs Actual Glucose



Residuals vs Predicted



Streamlit Dashboard

Page 1 — Raw vs clean data side-by-side + zero counts bar chart.

Page 2 — Correlation heatmap, Glucose distribution, feature scatter plots.

Page 3 — Predicted vs Actual, Residuals plot, Coefficient bar chart, metrics.

Page 4 — Interactive predictor: adjust patient sliders → real-time Glucose prediction.

✓ Normal <100 mg/dL



Pre-diabetic 100–125

! Diabetic ≥ 126 mg/dL

Conclusion

Pipeline completed — Preprocessing, EDA, modeling, and deployment done end-to-end.

R² of 0.33 — Reasonable for a linear model; Glucose is influenced by external factors not in this dataset.

Interpretability — Coefficients clearly show which features drive Glucose.

Future work — Random Forest or Gradient Boosting would capture non-linear relationships and push R² significantly higher.